PHYSICS MASTER SYLLABUS

CLASS 11 • NCERT / CBSE

Chapter 1: Physical World

- Scope of Physics
- Physics and Technology
- Fundamental Forces
- Nature of Physical Laws

Chapter 2: Units and Measurements

- Physical Quantities
- SI System of Units
- Fundamental Units
- Derived Units
- Measurement of Length
- Measurement of Mass
- Measurement of Time
- Accuracy
- Precision
- Significant Figures
- Errors in Measurement
- Dimensions
- Dimensional Formulae
- Dimensional Analysis

Chapter 3: Motion in a Straight Line

- Position
- Path Length
- Displacement
- Speed
- Velocity
- Average Velocity
- Instantaneous Velocity
- Acceleration
- Equations of Motion
- Motion Graphs

Chapter 4: Motion in a Plane

- Scalars
- Vectors
- Vector Addition
- Resolution of Vectors
- Relative Velocity
- Projectile Motion
- Uniform Circular Motion

Chapter 5: Laws of Motion

- Force
- Inertia
- Newton's First Law
- Newton's Second Law

- Newton's Third Law
- Momentum
- Impulse
- Friction
- Circular Motion Dynamics

Chapter 6: Work, Energy and Power

- Work
- Kinetic Energy
- Work-Energy Theorem
- Potential Energy
- Conservation of Mechanical Energy
- Power
- Elastic Collision
- Inelastic Collision

Chapter 7: System of Particles and Rotational Motion

- Centre of Mass
- Motion of Centre of Mass
- Linear Momentum
- Angular Displacement
- Angular Velocity
- Angular Acceleration
- Torque
- Angular Momentum
- Moment of Inertia
- Radius of Gyration
- Rotational Kinetic Energy
- Rolling Motion
- Equilibrium of Rigid Bodies

Chapter 8: Gravitation

- Universal Law of Gravitation
- Gravitational Constant
- Gravitational Field
- Acceleration Due to Gravity
- Gravitational Potential
- Gravitational Potential Energy
- Escape Speed
- Orbital Motion
- Satellites

Chapter 9: Mechanical Properties of Solids

- Elastic Behaviour
- Stress
- Strain
- Hooke's Law
- Young's Modulus
- Bulk Modulus
- Shear Modulus

Chapter 10: Mechanical Properties of Fluids

- Pressure
- Fluid Pressure
- Pascal's Law
- Hydraulic Machines
- Buoyancy
- Archimedes' Principle
- Surface Tension
- Capillarity
- Viscosity
- Stokes' Law
- Bernoulli's Principle

Chapter 11: Thermal Properties of Matter

- Temperature
- Heat
- Thermal Expansion
- Specific Heat Capacity
- Calorimetry
- Heat Transfer
- Conduction
- Convection
- Radiation
- Newton's Law of Cooling

Chapter 12: Thermodynamics

- Thermal Equilibrium
- Thermodynamic System
- Internal Energy
- Heat
- Work
- First Law of Thermodynamics
- Thermodynamic Processes
- Second Law of Thermodynamics
- Heat Engine
- Refrigerator

Chapter 13: Kinetic Theory

- Molecular Nature of Matter
- Ideal Gas
- Gas Laws
- Kinetic Theory of Gases
- Degrees of Freedom
- Law of Equipartition of Energy
- Mean Free Path

Chapter 14: Oscillations

- Periodic Motion
- Oscillatory Motion
- Simple Harmonic Motion
- Displacement
- Velocity
- Acceleration

- Energy in SHM
- Damped Oscillations
- Forced Oscillations
- Resonance

Chapter 15: Waves

- Wave Motion
- Mechanical Waves
- Longitudinal Waves
- Transverse Waves
- Wave Parameters
- Wave Speed
- Superposition Principle
- Standing Waves
- Beats
- Doppler Effect